# CS4013 Software Development Project
# University of Limerick
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**Objectives:**

- Apply an object-oriented approach to the design and development of a software application in Java.
- Apply the concepts and techniques introduced in lectures, tutorials and labs to the design and development of a small software application.

## UL Student Records Systems

**System Requirements:**

**The University of Limerick has updated its Handbook of Academic Regulations and Procedures for AY26/27 (**Handbook of Academic Regulations 2026/27**).**

**Chapters 2 and 3 in particular focus on the rules around grading, student progression etc.. These are the rules which are implemented within the Student Records System.**

You are required to build a student records system for UL (in so far as you can). UL offers many different types of programmes, which are of varying duration, at varying levels such as undergraduate, postgraduate and also include both taught programmes and research programmes. Programmes are run by Departments/Schools and these departments hold examination boards at the end of each review period to review student progression. Students may be requested to repeat a module, link-in to modules, repeat semester, year etc…

Your system should store student results (grades in modules each semester) and should perform QCA calculations per semester/year/programme (and others as deemed necessary) for the different types of programmes available in UL. The system should be able to identify students that do not meet the minimum academic standards for progression.

Students should be able to review their own transcript. Faculty should be able to look up transcripts. Faculty should be able to submit results for their modules each year/semester. Departments should be able to hold exam boards to review progression.

All files associated with this application should be stored as csv files and a command line interface should be included.

Core Set of required features:

- Add student(s)
- Add programme
- Add module
- Register student on programme
- Record module results

- Calculate semester QCA
- Calculate overall QCA
- Generate transcript
- Identify progression status

Sample additional features:

- Repeat decisions
- Compensation rules
- Link-in modules
- Awards classification

The design should demonstrate appropriate use of core object-oriented principles including encapsulation, inheritance, polymorphism and abstraction where suitable.

It is not expected that every regulation will be implemented. Each group should implement a representative subset of progression and award rules and clearly document the assumptions made.

A GitHub repository should be used to support team collaboration and manage code submissions. Commit history should demonstrate sustained development across the project duration. A large volume of code committed in one commit or immediately prior to submission may be considered when assessing contributions.

**Suggestions**

- In terms of managing the project, consider adopting some of the ideas from an Agile approach such as Scrum…for example such an approach advocates daily meetings. In the context of this project and your work on other modules, maybe you could agree to meet every 2-3 days where it is made clear what each team member will work on.

**Project Deliverables**

This project must be implemented in the Java programming language. As can be seen from the deliverables below this project is not just an implementation project, it also involves design, documentation and testing.

The following are the deliverables for the project:

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- A document outlining the Class Responsibility Collaboration (CRC) cards. These can be created in a word document.
- A UML class diagram showing the relationships between the classes.
- Documentation for the software generated using the Javadoc utility and a help file describing briefly how to run the application.

- The source code for the system where each Java class is stored in a separate file.
- Any CSV files that are required by the system should also be included. These CSV files must be populated with test data to allow full execution and evaluation of the system during the project interview.
- The use of a GitHub repository for source code version management.
- A completed Contributions document which will be circulated separately on Brightspace.
- A completed Generative AI declaration document (following the guidelines in the Generative AI Usage policy which will be circulated separately on Brightspace.

**Rules and Regulations**

- You must undertake this assignment on a group basis. Strictly groups with max size 4.
- You must add your details to the project group spreadsheet by the deadline specified (link to spreadsheet will be provided via a Brightspace announcement, deadline for this also provided in this communication). Failure to do so will result in zero marks being awarded for the project.
- As this is a software development module, all team members are expected to make a substantial contribution to the implementation of the system. Contributions limited primarily to documentation, testing, project management, or presentation activities will not normally be considered sufficient to achieve the learning outcomes of the module.
- ***You will be required to conduct an interview where each team member will be requested to give a code and documentation walkthrough, demo and answer any related OO questions of the artifacts submitted as part of the project***. *Failure of (an) individual(s) to attend the interview (without proper explanation/evidence e.g. Doctor's Cert) will result in 0 marks being awarded.*
- The project will be completed in groups; however, final marks may be adjusted at the discretion of the evaluator(s) to reflect individual contribution and understanding. Evidence may include project interviews, repository history, project artefacts, peer feedback, and the ability to explain and justify design and implementation decisions. All team members are expected to contribute substantially to each part of the project.
- Every student in a group is expected to have a satisfactory understanding of the submitted system, its design and the contributions they made to its development.
- *The interview will take place at a selected time from the start of week 13 to the end of the examination period.* It is your responsibility to prepare adequately for such a demonstration and to be knowledgeable about the design, implementation and documentation submitted. Students may be asked to explain, modify, or justify any submitted artefact during the project interview.
- You must abide by the guidelines outlined in the Generative AI Usage Policy (circulated separately).
- Failure to satisfactorily demonstrate your project will result in an F grade for the assignment irrespective of the contents of the submitted documentation. An unsatisfactory demonstration is also likely to be considered plagiarism. It is also

your responsibility to ensure that other students do not submit any part of your work as part of their project. This will also be deemed as plagiarism. Plagarism will result in an F-grade being awarded for all in-term assessments and the University Code of Conduct will be applied.

- An F-grade will be awarded to all team members if the contributions file is not included, complete or if an inaccurate contributions file is submitted.
- The project must be submitted as a single compressed file to Brightspace by the stated deadline. Submission deadline is **12h00 Monday Week 13.** A **penalty** of 10% for each day thereafter will be imposed for late submissions. Submissions will not be accepted after 12h00 Thursday Week 13.
- Accidental loss of work will not be accepted as an excuse.

**Grading Rubric**

The following grading rubric will apply:

| Assessment Area | Marks |
|---|---|
| Analysis and OO Design | 20 |
| OO Implementation and Functional Requirements | 50 |
| Documentation | 10 |
| Testing | 10 |
| Interview/Demo | 10 |

As outlined above under Rules and Regulations, group marks may be adjusted to reflect individual contribution and understanding.

**Version History:**

Version 1.0 released Friday 30th Oct. 2026.